Programming Reference: Java Fundamentals

Educational Series

December 29, 2025

1 Introduction to Java

Java is a high-level, class-based, object-oriented programming language designed to have as few implementation dependencies as possible. Its famous philosophy is **"Write Once, Run Anywhere" (WORA)**.

1.1 Key Components

- **JDK (Java Development Kit):** The environment used to develop Java applications.
- **JRE (Java Runtime Environment):** Provides the libraries and JVM required to run Java programs.
- **JVM (Java Virtual Machine):** An engine that provides a runtime environment to drive the Java Code (Bytecode).

2 Access Modifiers

Access modifiers define the scope and visibility of a class, method, or variable:

- **Public:** Accessible from any other class.
- **Private:** Accessible only within the same class.
- **Protected:** Accessible within the same package and by subclasses.
- **Default:** Accessible only within the same package (no keyword used).

3 Object-Oriented Programming (OOP)

Java is built around four pillars:

1. **Encapsulation:** Using private variables and public *getters/setters*.
2. **Inheritance:** Using the `extends` keyword to inherit from a parent class.
3. **Polymorphism:** Method Overloading (same name, different params) and Overriding (same name/params in a subclass).
4. **Abstraction:** Using **Abstract Classes** or **Interfaces**.

3.1 Interface vs. Abstract Class

- **Interface:** Used for *multiple inheritance*; all methods are implicitly abstract (until Java 8). Use `implements`.
- **Abstract Class:** Can have both abstract and concrete methods. Use `extends`.

4 Exception Handling

Exceptions are events that disrupt the normal flow of a program. Java uses the `try-catch-finally` block:

```java
try {
    int data = 100 / 0; // Throws ArithmeticException
} catch (ArithmeticException e) {
    System.out.println("Cannot divide by zero!");
} finally {
    System.out.println("This block always executes.");
}
```

5 Java Collections Framework

Used to store and manipulate a group of objects.

- **List:** Ordered collection (e.g., `ArrayList`, `LinkedList`).
- **Set:** Collection of unique elements (e.g., `HashSet`).
- **Map:** Key-Value pairs (e.g., `HashMap`).

6 The "static" and "final" Keywords

- **static:** Belongs to the class rather than instances. Shared by all objects.
- **final:** Used to define constants (variables that can't change), methods that can't be overridden, or classes that can't be inherited.

7 Memory Management

Java features **Automatic Garbage Collection**, which automatically deletes unused objects to free up memory.